Ferns and Petals (FNP) Sales Analysis Project Report

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Project Type: Data Analytics Dashboard Project
Tools Used: Microsoft Excel, Pivot Tables, Data Visualization

1. Introduction

Ferns and Petals (FNP) is a gifting platform that delivers products for various occasions such as Valentine's Day, Diwali, Raksha Bandhan, Birthdays, and Anniversaries. The company receives a large number of customer orders across multiple cities and product categories. Understanding sales trends, customer purchasing behavior, and product performance is essential for improving business strategies and customer satisfaction.

This project analyzes transactional sales data to uncover patterns in revenue, product popularity, customer spending behavior, and delivery efficiency. The insights generated from this analysis can help the company optimize marketing campaigns, improve logistics, and increase overall sales.

2. Problem Statement

The company wants to analyze its sales data to understand business performance and identify key opportunities for growth. The main objectives of the analysis include:

- Identify total revenue generated from sales.
- Analyze average order delivery time.
- Understand monthly sales performance.
- Determine the top-performing products by revenue.
- Analyze average customer spending per order.
- Identify the top cities placing the highest number of orders.
- Compare revenue generated across different occasions.
- Understand product popularity during different events.

3. Dataset Description

The dataset contains transactional records related to customer orders. Each record represents a purchase made by a customer and includes information about the product, order date, delivery date, city, and occasion.

- Order ID – Unique identifier for each order
- Product Name – Name of the product purchased
- Product Category – Category of the product
- Occasion – Event associated with the purchase
- City – Customer location
- Order Date – Date when the order was placed
- Delivery Date – Date when the order was delivered
- Revenue – Total value of the order
- Quantity – Number of items purchased

4. Data Cleaning and Preparation

Before performing analysis, the dataset was cleaned and prepared to ensure accuracy and consistency. The following steps were performed:

- Checked for missing or inconsistent values in important columns.
- Converted order and delivery dates into standard date formats.
- Calculated delivery time by subtracting order date from delivery date.
- Removed duplicate records to maintain data integrity.
- Standardized product categories and occasion names.

5. Data Analysis Techniques

Microsoft Excel was used to perform the analysis. Pivot tables and charts were used to summarize and visualize the data. The following analytical techniques were applied:

- Revenue analysis across different occasions
- Monthly sales trend analysis
- Top products by revenue identification
- City-wise order distribution
- Customer spending analysis
- Delivery time analysis

6. Dashboard Development

An interactive dashboard was developed using Excel to visualize key performance indicators and business insights. The dashboard includes charts and KPI metrics such as:

- Total Orders
- Total Revenue
- Average Delivery Time
- Average Customer Spending
- Monthly Revenue Trends

- Top Products by Revenue
- Top Cities by Orders

7. Key Insights

The analysis of the dataset revealed several important insights about the company's sales performance:

- Occasion-based events such as Raksha Bandhan and anniversaries generate the highest revenue.
- Sales show seasonal patterns with peaks during festival months.
- Certain product categories such as gift sets, sweets, and soft toys contribute significantly to revenue.
- Customers tend to purchase premium gift combinations, resulting in higher average order values.
- Some cities consistently generate a higher number of orders compared to others.

8. Business Recommendations

Based on the insights generated from the analysis, the following strategies are recommended:

- Increase marketing campaigns during major gifting occasions.
- Promote high-performing products to maximize revenue.
- Target high-demand cities with localized promotions.
- Improve delivery efficiency to enhance customer satisfaction.

9. Conclusion

This project demonstrates how sales data can be transformed into meaningful business insights through data analysis and visualization. The dashboard provides a comprehensive overview of sales performance and helps stakeholders identify trends, opportunities, and areas for improvement. By leveraging these insights, the company can make informed decisions to improve marketing strategies, optimize product offerings, and enhance customer experience.